Questions

Q1.

Figure 20 shows a magnet and a coil.

The coil is connected to a sensitive centre-zero ammeter.

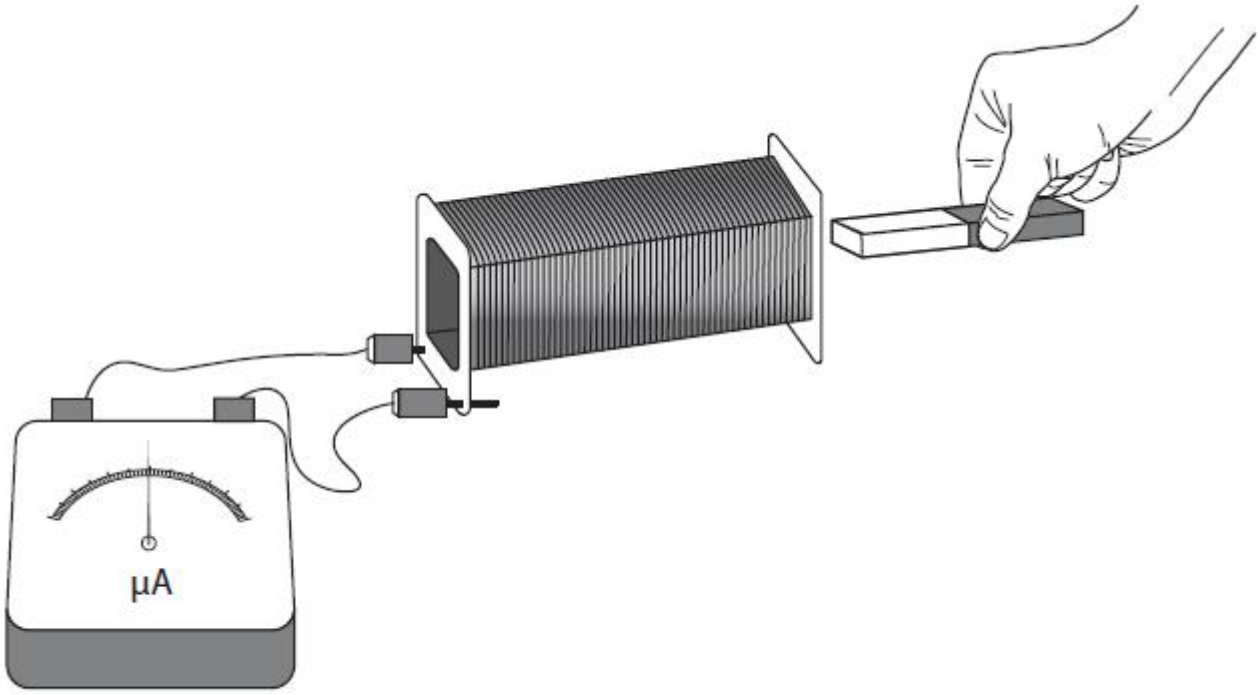

Figure 20

Explain what will be observed on the meter when the magnet is pushed in and pulled out of the coil, repeatedly.

(3)

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(Total for question = 3 marks)

Q2.

Figure 22 shows how electricity is delivered efficiently from a power station (P) to homes (T).

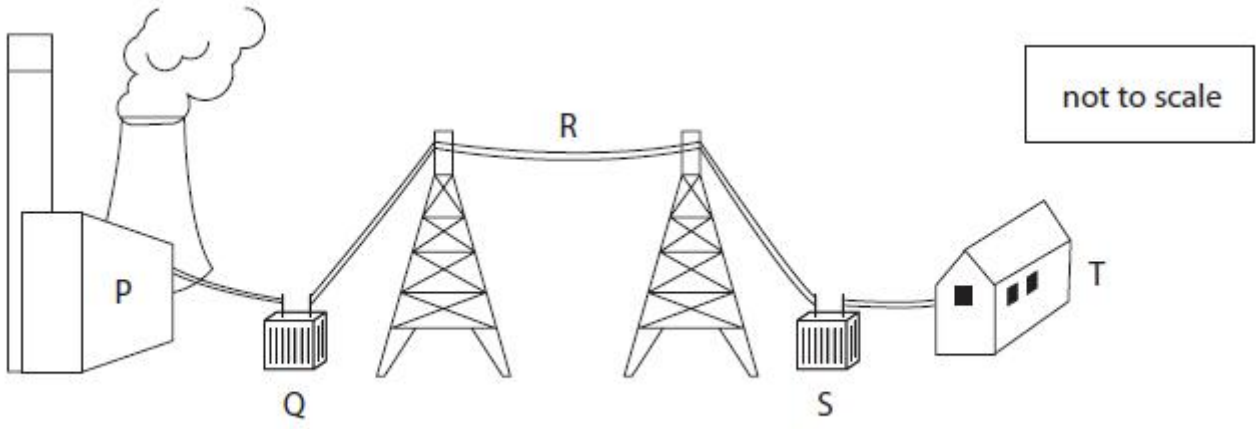

Figure 22

Using Figure 22, explain the stages in the process of delivering electricity efficiently from P to T.

Your answer should include details of the effects that Q, R and S have on efficiency.

(6)

(Total for question = 6 marks)

Q3.

Figure 17 is a diagram representing a loudspeaker.

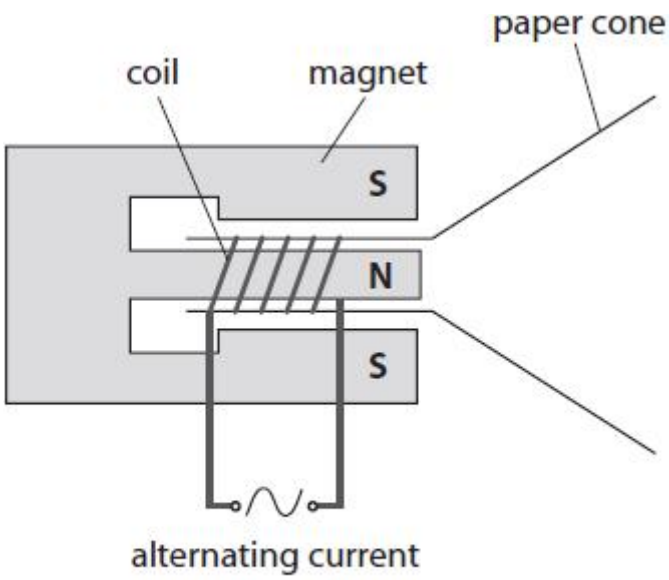

Figure 17

Explain how sound is produced when an alternating current is supplied to the coil of the loudspeaker.

(4)

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(Total for question = 4 marks)

Q4.

A transformer converts a voltage of 11 000 V to 230 V.

Explain how the design of this transformer enables the voltage to be converted from 11 000 V to 230 V.

Your answer should include

- details of the structure of a transformer
- how a transformer works, using ideas of electromagnetic induction
- how the design of this transformer enables this exact voltage of 230 V to be obtained.

You may draw a diagram to help your answer.

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(Total for question = 6 marks)

Q5.

The battery inside an electric toothbrush is recharged when the toothbrush is placed on its charger.

Figure 9 shows part of the toothbrush and the charger.

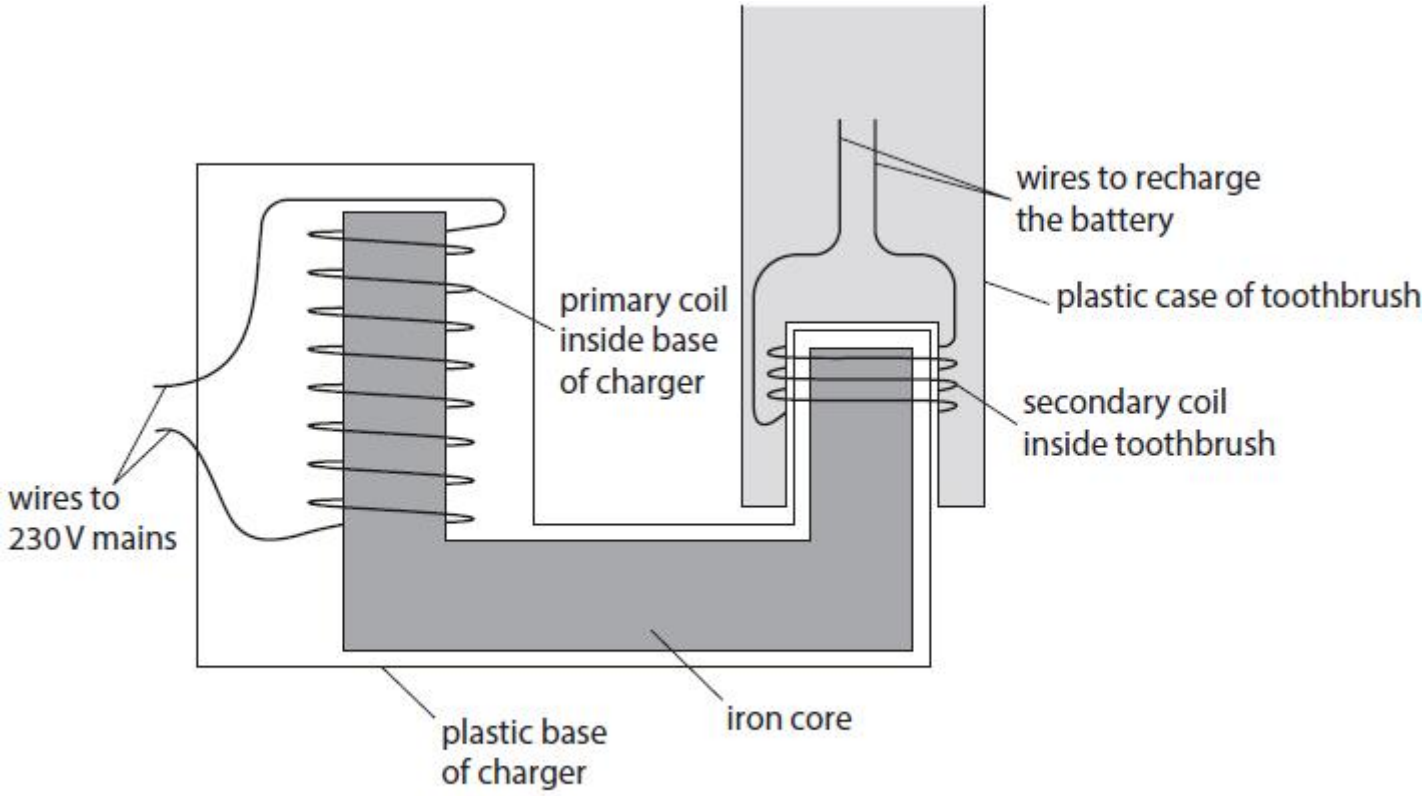

Figure 9

(i) Explain how the current is produced in the secondary coil of the transformer in Figure 9.

(3)

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(ii) The potential difference across the primary coil is 230 V.

The potential difference across the secondary coil is 1.5 V.
The primary coil has 1840 turns of wire.
Calculate the number of turns of wire in the secondary coil.

(2)

number of turns = ...

(Total for question = 5 marks)

Q6.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

There is an alternating current of 3 A in the primary coil of a transformer.

There is an alternating current of 6 A in the secondary coil of the transformer.

The transformer is 100% efficient.

(i) The size of the potential difference (voltage) across the secondary coil is

(1)

☐ **A** twice the size of the current in the primary coil
☐ **B** half the size of the current in the primary coil
☐ **C** twice the size of the voltage across the primary coil
☐ **D** half the size of the voltage across the primary coil

(ii) Explain how an alternating current in the primary coil causes an alternating current in the secondary coil of the transformer.

(3)

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(Total for question = 4 marks)

Q7.

Figure 23 shows a magnet being dropped through a coil.

The coil is connected to a sensitive voltmeter.

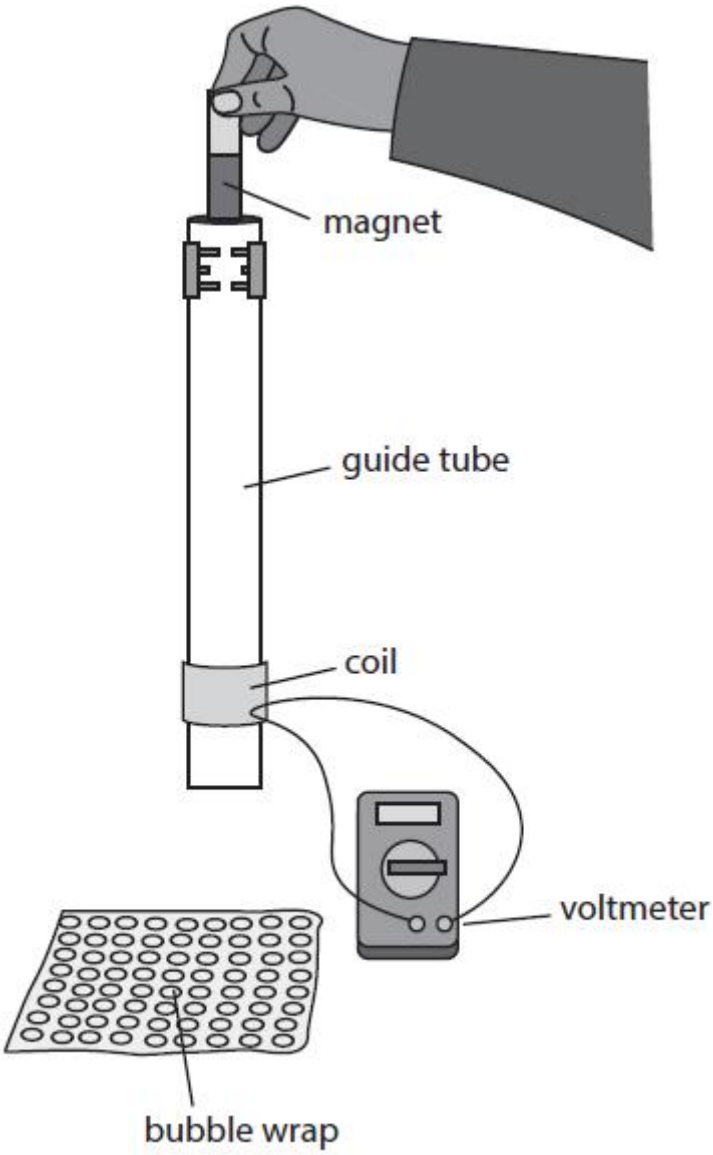

Figure 23

(i) Explain why the voltmeter shows a reading as the magnet passes through the coil.

(3)

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(ii) A student develops this investigation to find out how the reading on the voltmeter depends on the height the magnet is dropped from.

Describe how the student could use the arrangement in Figure 23 to do this investigation.

(4)

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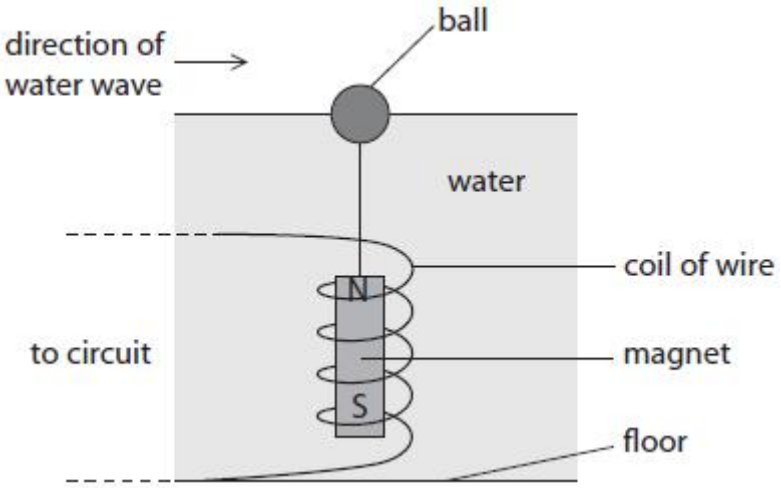
(Total for question = 7 marks)

Q8.

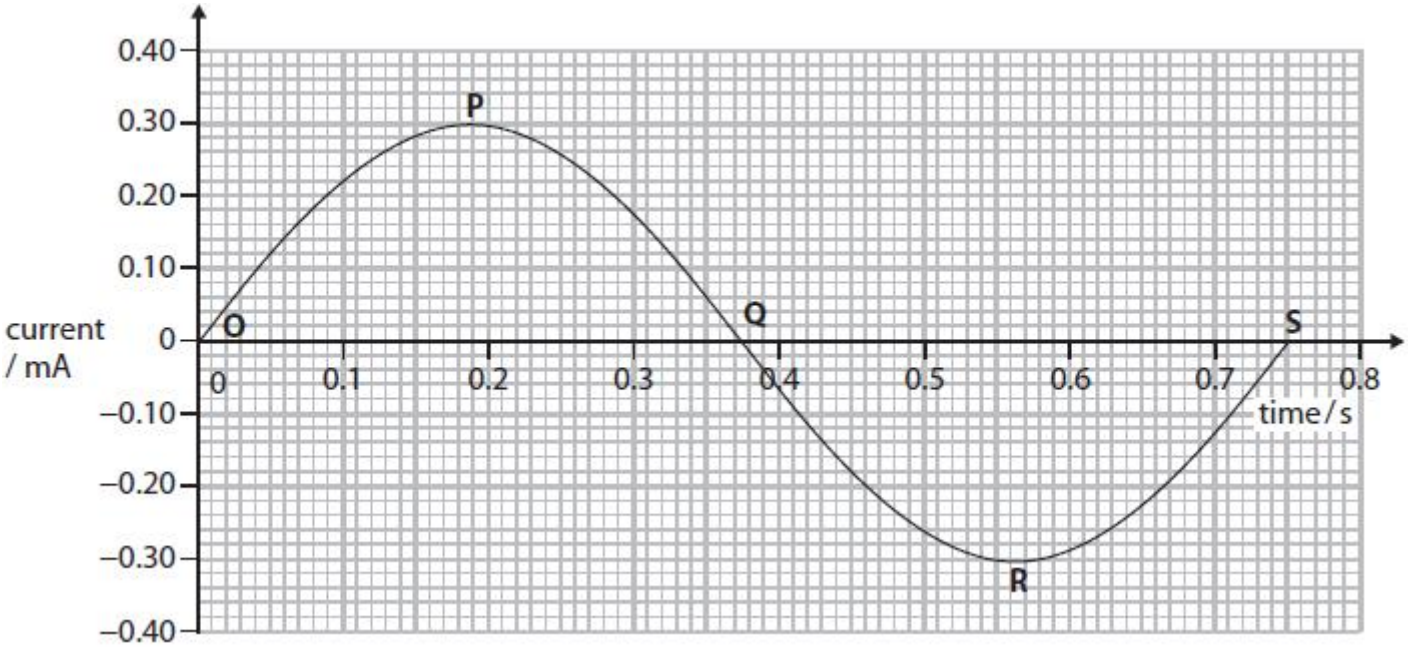
* The diagram shows a model used to generate electricity from water waves in a tank.

A ball floats on the surface of the water in the tank.
A coil of wire is fixed to the floor of the tank.
A magnet is suspended from the ball inside the coil.

When a wave is sent along the surface of the water the ball moves up and down.



The graph shows the current induced in the coil.



Explain how this current is induced in the coil in the model.
You should refer to the model and to the labelled points on the graph in your answer.

(6)

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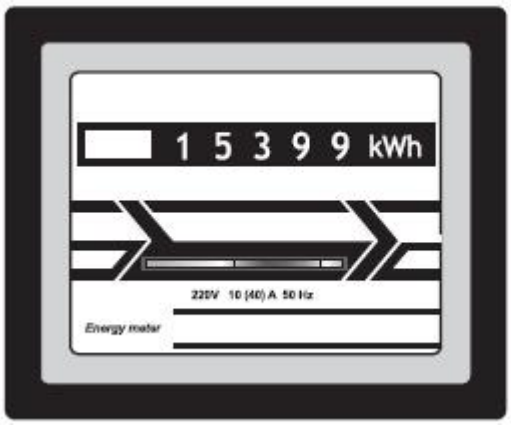
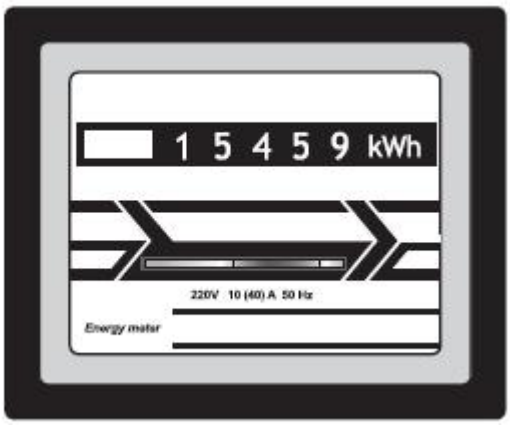
Q9.

(a) Electricity costs 20p for each kW h.
The pictures show a domestic electricity meter at two different times.





(i) Calculate the cost of the electricity used between the two readings.

(2)

cost =.....p

(ii) The time between these two readings is 15 hours.
Calculate the average power supplied.

(2)

average power =.....kW

(b) Explain why step-up transformers are used in the transmission of electricity in the National Grid.

(2)

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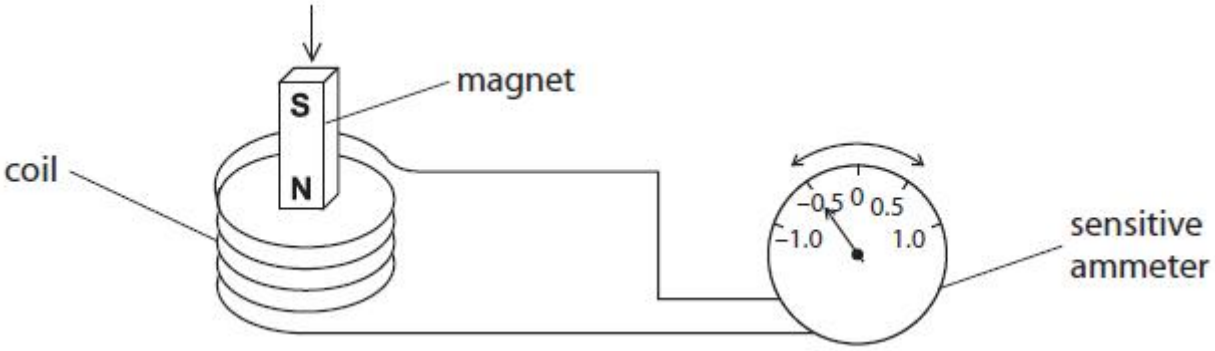
*(c) The diagram shows a magnet moving into a coil of wire.

The coil of wire is attached to a sensitive ammeter.



The moving magnet and the coil of wire are producing an electric current.

The size and direction of the current can be changed in a number of ways.

Describe changes that can be made to produce different currents and the effect of each change.

(6)

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(Total for Question = 12 marks)

Q10.

* Figure 19 shows a coil of wire that is being rotated between the poles of a magnet.

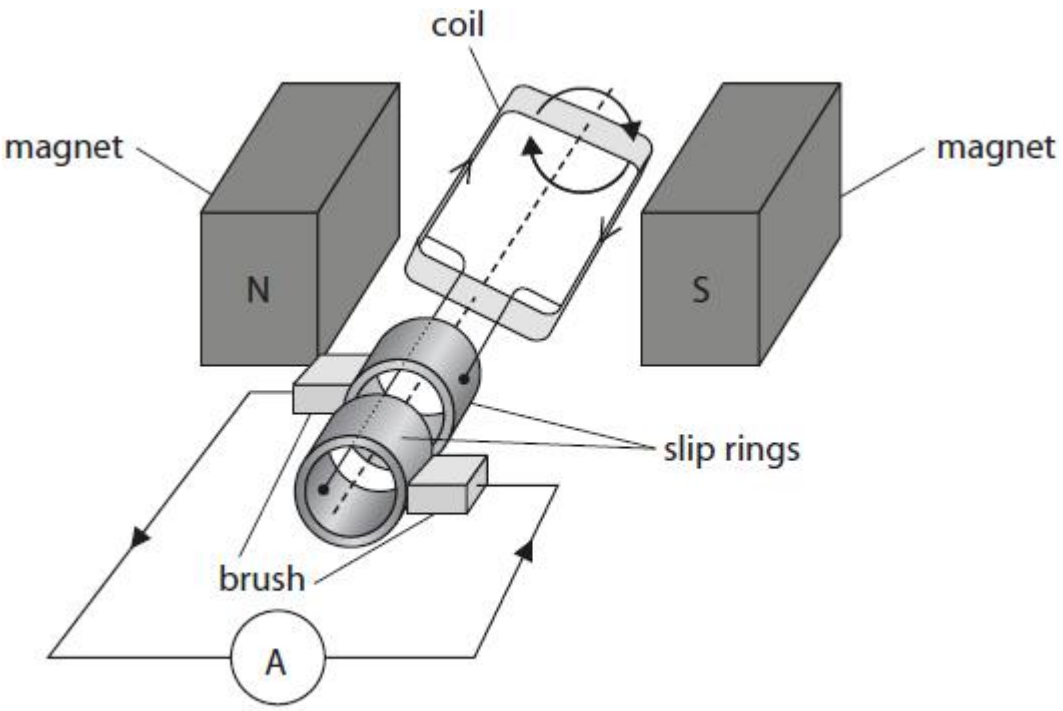

Figure 19

Figure 20 shows how the current in the coil changes during **one** complete rotation of the coil.

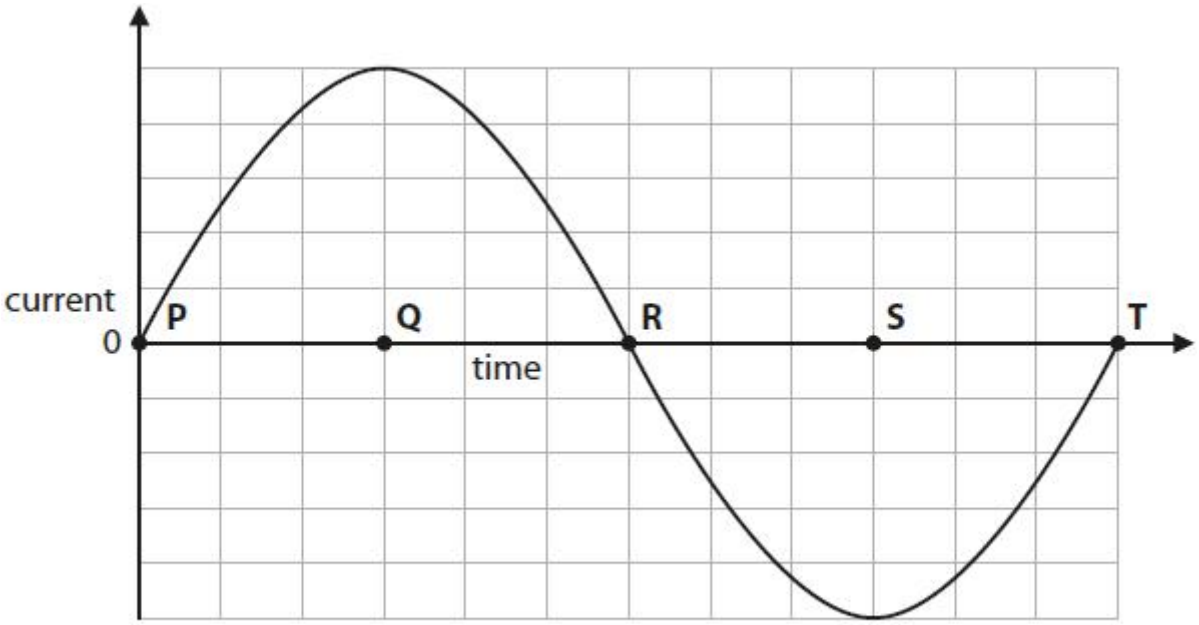

Figure 20

Explain why the current changes in the way shown by the graph in Figure 20.

Your answer should include details of the position of the coil relative to the magnet at each of the times labelled P, Q, R, S and T.

You may use diagrams to help your answer.

(6)

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(Total for question = 6 marks)

Q11.

(a) The picture shows a wind-powered generator used to produce electricity for a house.

© The Wall Street Journal

(i) The table shows some electrical components.
Put ticks in the table next to the **two** components the generator must contain.

(2)

component	✓
ammeter	
coil of wire	
battery	
magnet	
voltmeter	

(ii) Explain why the voltage produced by this wind-powered generator is not always the same.

(2)

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(iii) It would cost the homeowner 15 p to buy 1.0 kW h of electrical energy from the National Grid.

His generator has a maximum power of 2.0 kW.
The generator produces energy at this maximum power for 3 hours.
Calculate how much it would cost to buy the same amount of energy from the National Grid.

(2)

cost = ... p

(b) An electric kettle is plugged into a 230 V mains supply.
It has a power of 2.5 kW.

Use this equation to calculate the current in the kettle.

$$\text{current (in amps)} = \frac{\text{power (in watts)}}{\text{voltage (in volts)}}$$

(3)

current = ... A

(c) Suggest why a 2 kW wind-powered generator may not supply all the electrical energy needed in a house.

(1)

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Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
	an explanation linking **three** from:		**(3) AO1.2**
	needle oscillates (1)	needle vibrates / moves side to side	
	either side of (centre) zero (1)	positive and negative	
	(in response to) pole entering and pole leaving (end of coil) (1)	accept N / north / S / south for pole do not accept magnet	
	(producing) {p.d. / voltage / emf} induced (via changing magnetic field) (1)	accept current produced / induced	
	(producing) an alternating current (1)	accept (induce) an alternating emf / voltage	

Q2.

SSQ NO:	CS NO:	Answer	Mark
*		Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO1 strand 1 (6 marks)** • Q is a step-up transformer • step up V causes I to be lower • voltage increases (25 kV to 400 kV) • R is a transmission line / (national) grid /cable • smaller currents in transmission lines • less energy lost though heating those wires • V = I x R • smaller voltage drop across the transmission line • S is a step-down transformer reducing voltage to 230V • ready for use in homes T • detail of transformers – iron core + coils • transformers are not 100% efficient • idea of power as V x I or $P = I^2R$	**(6)** **AO1.1**

Level	Mark	Descriptor
	0	No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which may be inaccurate. Understanding of scientific ideas lacks detail. (AO1) • Presents an explanation with some structure and coherence. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1)

Summary for guidance

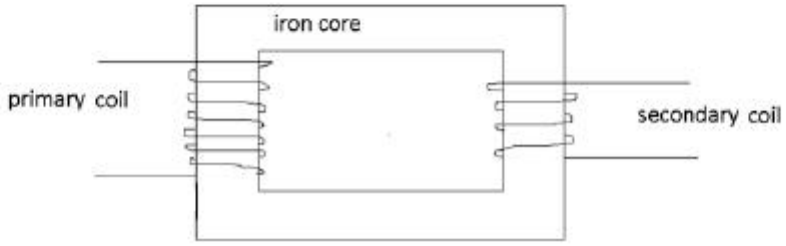
Level	Mark	Additional Guidance	General additional guidance – the decision within levels Eg - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance isolated ideas e.g. identifying two of Q, S and R	Possible candidate responses Q and S are transformers R is a wire / cable

Level 2	3–4	Additional guidance more detail about the process of what at least two of Q, R and S do / achieve	Possible candidate responses Q is a step-up transformer - voltage increases R is a high voltage transmission line / cable / part of the National Grid S is a step-down transformer → idea of reducing voltage to 230V
Level 3	5–6	Additional guidance understanding is detailed and fully developed. includes detail about functions and efficiency explanation	Possible candidate responses need for step up and step-down functions via transformers to transfer energy at high voltages (voltage may be specified e.g. 400kV) transformers are not 100% efficient smaller currents in transmission lines so less energy lost though heating those wires: makes system more efficient

Q3.

Question number	Answer	Additional guidance	Mark
(b)	an explanation linking in a logical order any four of the following:- (alternating) current produces (changing) magnetic field (around coil)(1) the coil is in a magnetic field (of fixed magnets) (1) (varying current in magnetic field) produces a force (1) the force on the coil /cone (continuously) changes direction (1) the paper cone /coil vibrates/ moves to and fro (1)	 magnetic fields interact making the air molecules (in the cone) vibrate	**(4)**

Q4.

SSQ NO:	CS NO:	Answer	Mark
*		Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO1 strand 1 (6 marks)** **Elements of physics understanding included in explaining the design:** **details of structure**  	(6) Exp AO1.1
		credit diagram for **construction** as part of the explanation • (closed) iron core • primary and secondary coils • primary coils number (considerably) different from secondary coils number **how a transformer works** • a.c. in primary • causes changing magnetic field in (iron) core • induces a.c. voltage in secondary coil	

		• stepping down or reducing the voltage / more turns on primary than secondary **how 230V is obtained** • achieved by same ratios of $\frac{V_P}{V_S}$ and $\frac{N_P}{N_S}$ • ratio 11 000:230 i.e. ≈ 48:1	

Level	Mark	Descriptor
	0	No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which may be inaccurate. Understanding of scientific ideas lacks detail. (AO1) • Presents an explanation with some structure and coherence. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1)

Level	Mark	Additional Guidance	General additional guidance – the decision within levels Eg - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance basic ideas about construction OR how transformer works OR use of ratio equation	Possible candidate responses use of primary and secondary coils on iron core use of a.c. / iron core and (changing) magnetic field step down transformer with smaller number of turns for secondary
Level 2	3–4	Additional guidance more developed ideas about construction AND how transformer works OR use of ratio idea	Possible candidate responses use of different sized primary and secondary coils and iron core use of a.c. in primary which causes changing magnetic field in (iron) core use of $\frac{V_p}{V_s} = \frac{N_p}{N_s}$

Level 3	5–6	Additional guidance understanding is detailed and fully developed. includes detail about construction **AND** how transformer works **AND** use of ratio idea for transformers one aspect of the three treated weakly may be tolerated	Possible candidate responses use of primary and secondary coils and iron core with primary coils number (considerably) bigger than secondary coils number AND (alternating) current in primary coil produces changing magnetic field in (iron) core inducing an a.c. voltage in the secondary coil AND ratio is 11000 / 230 or 48:1

Q5.

Question number	Answer	Additional guidance	Mark
(i)	an explanation linking alternating current in the primary coil (1) changing magnetic field (in the iron core) (1) induces an (alternating) p.d. (across the secondary coil) (1)	 mains is alternating / a.c. induces (alternating) current (in the secondary coil) ignore step-up/step-down	(3) AO1

Question number	Answer	Additional guidance	Mark
(ii)	substitution into $\frac{V_p}{V_s} = \frac{N_p}{N_s}$ (1) $\frac{230}{1.5} = \frac{1840}{N_s}$ rearrangement and evaluation (1) (N_s =) 12 **OR ALTERNATIVE METHOD** rearrangement (1) (N_s =) $\frac{V_s \times N_p}{V_p}$ substitution and evaluation (1) (N_s =) 12	 (N_s =) $\frac{1840 \times 1.5}{230}$ award full marks for correct answer without working.	(2) AO2

Q6.

Question number	Answer	Additional guidance	Mark
i	D half the size of the voltage across the primary coil A and B are incorrect because the voltage will not necessarily be twice or half the value of the current C is incorrect because the voltage across secondary coil will be less than that across the primary coil		**(1)** **AO2**

Question number	Answer	Additional guidance	Mark
ii	an explanation linking **three** of magnetic field in primary / secondary coil / core (due to current) (1) magnetic field is alternating (1) (this magnetic) field cuts/links secondary coil (1) <u>induces</u> an alternating voltage (across secondary coil) (1)		**(3)** **AO1**

Q7.

Question number	Answer	Additional guidance	Mark
i	an explanation linking any **three** from (magnetic) field lines cut / intercepted by coil (1) **induces** a voltage / current (in the coil) (1) voltmeter measures (induced) voltage / potential difference (1) electrons (in the wire) experience a force / move in response to (a changing) field (1) (induced) voltage / current changes magnitude / direction as magnet passes through (1)	(magnetic) field lines interact with coil / act on coil coil experiences changing (magnetic) field / flux (linkage) produces magnetism or magnetic poles in the coil voltmeter reads / shows current voltmeter measures induced voltage scores 2	**(3)** **AO2.2**

Question number	Answer	Additional guidance	Mark
ii	practical suggestions to include any **four** from drop magnet from different heights (1) **use** of metre rule (1) change height in steps (e.g. of 5 cm) (1) **record** (maximum) meter reading / voltage (1) repeat readings for each drop **and** take an average (1) set the digital voltmeter on a.c. (1) plot an appropriate graph – e.g. voltmeter reading against height (1)	move the coil to different positions use different length tubes allow ruler / tape measure	**(4) exp AO3.3**

Q8.

Question Number		Indicative Content	Mark
QWC	*	An explanation including some of the following points • a current/voltage/emf is induced when there is relative movement between a magnet and a coil of wire • the current is bigger when the movement is faster • the current is alternating/regularly changing direction • the current is zero when the magnet is not moving • points P and R on the graph correspond to the fastest movement of the magnet • the magnet is changing direction at points O, Q, S on the graph (quoting positive and negative current values from graph is sufficient to indicate a change in direction of current on graph) • the magnet is at the top/bottom of its movement at points O, Q, S on the graph • the magnet is not moving at points O, Q, S on the graph IGNORE references to number of turns or stronger magnet	**(6)**

Level	0	No rewardable content
1	1 - 2	• a limited explanation linking induced current to idea of movement of magnet OR limited reference linking graph to type of current with no link to model e.g. magnet moving in coil (induces a current) / (magnetic) field lines cut coil OR (the graph shows) an alternating current • spelling, punctuation and grammar are used with limited accuracy • the answer communicates ideas using simple language and uses limited scientific terminology
2	3 - 4	• a simple explanation linking the motion of the magnet to the size/direction of the induced current OR {a limited explanation linking induced current to idea of movement of magnet AND limited reference linking graph to type of current with no link to model} e.g. Magnet moving in the coil induces a current. The faster it moves the bigger the induced current. OR Magnet moving in the coil induces a current. When the magnet changes direction, the current changes direction. OR Magnet moving in the coil induces a current. The graphs shows an alternating current. OR Magnet moving in the coil induces a current. The current is positive at P and negative at R. • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately
3	5 - 6	• a detailed explanation linking the motion of the magnet to the size/direction of the induced current AND reference to graph for one factor e.g. Magnet moving in the coil induces a current. The faster it moves the bigger the induced current. The magnet is moving fastest at point P on the graph. OR Magnet moving in the coil induces a current. When the magnet changes direction the current changes direction. At P and R the magnet is moving in opposite directions. OR Magnet moving in the coil induces a current. The current is positive at P and negative at R. The magnet is moving up at P and down at R. • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors

Q9.

Question Number	Answer	Acceptable answers	Mark
(a)(i)	60 (kW h/ units) (1) 60 x 20 (= 1200) (p) (1)	15459 - 15399 £12 ecf Award full marks for correct answer with no working £12 scores 2 Power of Ten error scores maximum 1 60 in answer space with no working scores 1	**(2)**

Question Number	Answer	Acceptable answers	Mark
(a)(ii)	60 / 15 (1) 4 (kW) (1)	Allow ecf from 6(a)(i) marking point 1 Award full marks for correct answer with no working	**(2)**

Question Number	Answer	Acceptable answers	Mark
(b)	An explanation linking any two of: • increase voltage (1) • decrease current (1) • reduce {loss / waste} of {energy / heat} (1)	Increase efficiency (of energy transmission) Ignore "more efficient" by itself Accept power instead of energy Accept no energy loss	**(2)**

Question Number		Indicative content	Mark
QWC	*** (c)**	A description to include some of the following points • speed of movement • stronger / more powerful (ORA) magnet • more turns / coils (ORA) • iron core • reversing movement • turning the magnet round • effect of any / each change • more conducting / less resistant wire • allow stronger current • allow ammeter reading / recording / voltage for current • allow moving coil Correct ideas but using inaccurate scientific terminology • larger / bigger magnet • more / longer movement Ignore • irrelevant information • speeds up current or more electricity	**(6)**
Level	**0**	no rewardable material	
1	**1-2**	• a limited description of any one change e.g. use more coils **OR** a stronger magnet. • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy	
2	**3-4**	• a simple description of any two different changes **OR** one change and its effect e.g. use more coils and a weaker magnet **OR** more coils more current • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy	
3	**5 - 6**	• a detailed description of a change linked to its effect and a second different change e.g. using more turns of wire makes a bigger current. Moving the magnet out. • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors	

Q10.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. • coil moving/cuts through magnetic field • coil experiences changing magnetic field • induces a voltage/current in the coil • size of voltage/current depends on rate of change of magnetic field • rate of change depends on angle between direction of movement and direction of field. • greatest (rate of) change when coil moving perpendicular to field. • maximum current at Q and S • coil is horizontal at Q and S • coil moving vertically up at Q and down at S • direction of current at Q opposite to S. • no change when coil moving parallel to field. • zero current at P, R and T • coil vertical at P, R, and T Credit can be given for correctly labelled diagrams	**(6)** **AO2 and AO3**

Level	Mark	Descriptor
	0	• No awardable content
Level 1	1–2	• Interpretation and evaluation of the information attempted but will be limited with a focus on mainly just one variable. Demonstrates limited synthesis of understanding. (AO3) • The explanation attempts to link and apply knowledge and understanding of scientific ideas, flawed or simplistic connections made between elements in the context of the question. (AO2)
Level 2	3–4	• Interpretation and evaluation of the information on both variables, synthesising mostly relevant understanding. (AO3) • The explanation is mostly supported through linkage and application of knowledge and understanding of scientific ideas, some logical connections made between elements in the context of the question. (AO2)
Level 3	5–6	• Interpretation and evaluation of the information, demonstrating throughout the skills of synthesising relevant understanding. (AO3) • The explanation is supported throughout by linkage and application of knowledge and understanding of scientific ideas, logical connections made between elements in the context of the question. (AO2)

Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance isolated facts about interaction of electric current and magnetic fields **or** one salient feature of the graph	Possible candidate responses the coil experiences a changing magnetic field as it rotates. Size of the (induced) current varies.
Level 2	3–4	Additional guidance simple description of why current changes (either in direction or magnitude) **and** reference to at least one relevant point on the graph.	Possible candidate responses at position R the (plane of the) coil is parallel to the field and there is no current **Or** at position Q the coil is moving quickly through the field and the current is large.
Level 3	5–6	Additional guidance Full description of why current changes in magnitude or direction **and** reference to at least two relevant points on the graph	Possible candidate responses At Q, the coil is horizontal and moving most quickly across the field so the current is at its greatest. At R the coils is vertical and moving parallel to the field so there is no current.

Q11.

Question Number	Answer	Acceptable answers	Mark
(a)(i)	**component** ☐ ammeter coil of wire ☐ battery magnet ☐ voltmeter	one mark for each correct tick deduct 1 mark for each extra tick	**(2)**

Question Number	Answer	Acceptable answers	Mark
(a)(ii)	Explanation linking any two of • wind (speed) is not constant (1) • voltage depends on wind speed (1)	need idea of varying wind {electrical energy / electricity} depends on wind speed higher wind speed gives {higher voltage/more electrical energy/more electricity} = 2 marks voltage is alternating = 2 marks	**(2)**

Question Number	Answer	Acceptable answers	Mark
(a)(iii)	(saving) = 2 × 3 × 15 (1) 90 (p) (1)	award full marks for correct answer with no working 2 × 3 × 0.15 (£) 0.90	**(2)**

Question Number	Answer	Acceptable answers	Mark
(b)	power = 2500 (W) (1) (current) = $\frac{2500}{230}$ (1) **ecf** 11 (A) (1)	award full marks for correct answer with no working [2.5/230 is 1 mark for these 2] 10.9 / 10.8... **accept** {0.01... / 0.11... / 1.1...} for 2 marks	**(3)**

Question Number	Answer	Acceptable answers	Mark
(c)	EITHER sometimes no / very little wind (1) OR some appliances rated above 2 kW (1)	need wind vague references to weather are insufficient may use more than one appliance at once **or** house needs more (than 2kW) power not enough power for kettle **ignore** references to electrical energy / electricity	**(1)**